

Unified System of Intentional Thermodynamics: Resolving the Quantum-Relativistic Paradox

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1. Abstract

This document presents the Unified System of Intentional Thermodynamics resolution to the fundamental conflict between quantum mechanics and general relativity. By redefining gravity as the quantized summation of mechanical work rather than geometric curvature, this framework provides a deterministic law governing the universe from discrete configurations to macroscopic aggregates.

2. The Resolution of Quantum Gravity

The "Holy Grail" paradox—the incompatibility of continuous gravity and discrete quantum particles—is resolved by identifying gravity as the collective, quantized summation of mechanical work.

- Gravity is identified as the intrinsic pressure gradient (P_g) generated by the rotational work of stable energy configurations.
- Because gravity is the energetic signature of the mass itself, it is inherently quantized at the micro-scale.
- This model eliminates the need for theoretical gravitons or the geometric distortion of spacetime, instead defining gravity as energy density ($E = P_g$) seeking equilibrium with the zero-energy vacuum baseline ($R_s = 0$).

3. Mathematical Formulary for Prediction

To test the framework's predictive power, the following equations are employed to calculate gravitational behavior across scales:

- **Unified Mechanical Law:** $P_g = \sum (\omega^2 \cdot \Omega_c)_i$.
- **Gravitational Constant (G_u):** $G_u = P_g / (\rho \cdot \Omega_c)$.

- **Stability Parity (Predictive Threshold):** $\forall p, i - \rho_{m, i} \leq 1.00e-43$.

4. Proposed Predictive Testing

The framework suggests that gravitational interactions can be predicted by calculating the energy-density (ρ) of discrete configurations.

- **Predicting Macro-Scale Gravity:** By summing the rotational work of constituent atoms, one can predict the total gravitational pressure of macroscopic bodies.
- **Stability Forecasting:** Utilizing the Stability Parity threshold, the framework can predict whether a specific configuration will maintain structural integrity or dissipate back to the vacuum baseline.
- **Simulation Validation:** The CAE V.7.0 validation confirms that discrete energy configurations aggregate into macro-scale fields with 100% stability, providing a reliable basis for forecasting field evolution.

5. Conclusion

By unifying discrete quantum work with aggregate macro-scale pressure, the Unified System of Intentional Thermodynamics provides a deterministic framework. This resolution bridges the physics of the very large and the very small, enabling precise calculations of gravitational interactions at any scale.

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